

EXTENDS *Naturals*

Symbolic *BB84* quantum states using *Dirac* notation

$QStates \triangleq \{ "|0\rangle", "|1\rangle", "|+\rangle", "|-\rangle" \}$
 $ZStates \triangleq \{ "|0\rangle", "|1\rangle" \}$
 $XStates \triangleq \{ "|+\rangle", "|-\rangle" \}$
 $Bases \triangleq \{ "Z", "X" \}$
 $Bits \triangleq \{ 0, 1 \}$
 $Basis(s) \triangleq \text{IF } s \in ZStates \text{ THEN "Z" ELSE "X"}$
 $BitValue(s) \triangleq \text{IF } s \in \{ "|0\rangle", "|+\rangle" \} \text{ THEN } 0 \text{ ELSE } 1$
 $StatesOfBasis(b) \triangleq \text{IF } b = "Z" \text{ THEN } ZStates \text{ ELSE } XStates$

VARIABLES *aliceState*, *channelState*, *bobBasis*, *bobResult*

vars $\triangleq \langle aliceState, channelState, bobBasis, bobResult \rangle$

$Init \triangleq \wedge aliceState \in QStates$
 $\wedge channelState = aliceState$
 $\wedge bobBasis \in Bases$
 $\wedge bobResult \in Bits$

Transmit models the quantum channel.

The qubit may remain unchanged (correct transmission) or change nondeterministically (noise or eavesdropping).

$Transmit \triangleq \wedge \exists b \in Bases : channelState' \in StatesOfBasis(b)$
 $\wedge \text{UNCHANGED } \langle aliceState, bobBasis, bobResult \rangle$

$Measure \triangleq \wedge \text{IF } Basis(channelState) = bobBasis$
 $\text{ THEN } bobResult' = BitValue(channelState)$
 $\text{ ELSE } bobResult' \in Bits$
 $\wedge \text{UNCHANGED } \langle aliceState, channelState, bobBasis \rangle$

Next $\triangleq Transmit \vee Measure$

Spec $\triangleq Init \wedge \Box [Next]_{vars}$

$$\begin{aligned}
Sifted &\triangleq Basis(aliceState) = bobBasis \\
AliceKeyBit &\triangleq \text{IF } Sifted \text{ THEN } BitValue(aliceState) \text{ ELSE "discard"} \\
BobKeyBit &\triangleq \text{IF } Sifted \text{ THEN } bobResult \text{ ELSE "discard"} \\
KeyAgreement &\triangleq Sifted \Rightarrow AliceKeyBit = BobKeyBit \\
Disturbance &\triangleq Sifted \wedge bobResult \neq BitValue(aliceState) \\
ChannelIntegrity &\triangleq channelState = aliceState \\
IdealCorrectness &\triangleq Basis(aliceState) = bobBasis \\
&\quad \Rightarrow bobResult = BitValue(aliceState) \\
PerfectChannelCorrectness &\triangleq ChannelIntegrity \Rightarrow IdealCorrectness \\
KeyTypeSafety &\triangleq AliceKeyBit \in Bits \cup \{\text{"discard"}\} \wedge BobKeyBit \in Bits \cup \{\text{"discard"}\}
\end{aligned}$$

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